

# Car Park API

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## Acknowledgement of Country

Transport for NSW acknowledges the traditional custodians of the land on which we work and live.

We pay our respects to Elders past and present and celebrate the diversity of Aboriginal people and their ongoing cultures and connections to the lands and waters of NSW.

Many of the transport routes we use today – from rail lines, to roads, to water crossings – follow the traditional Songlines, trade routes and ceremonial paths in Country that our nation's First Peoples followed for tens of thousands of years.

Transport for NSW is committed to honouring Aboriginal peoples' cultural and spiritual connections to the lands, waters and seas and their rich contribution to society.



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# Document control

|                             |                                                    |
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| <b>Superseded documents</b> | Car Park API Documentation v1.1                    |

## Versions

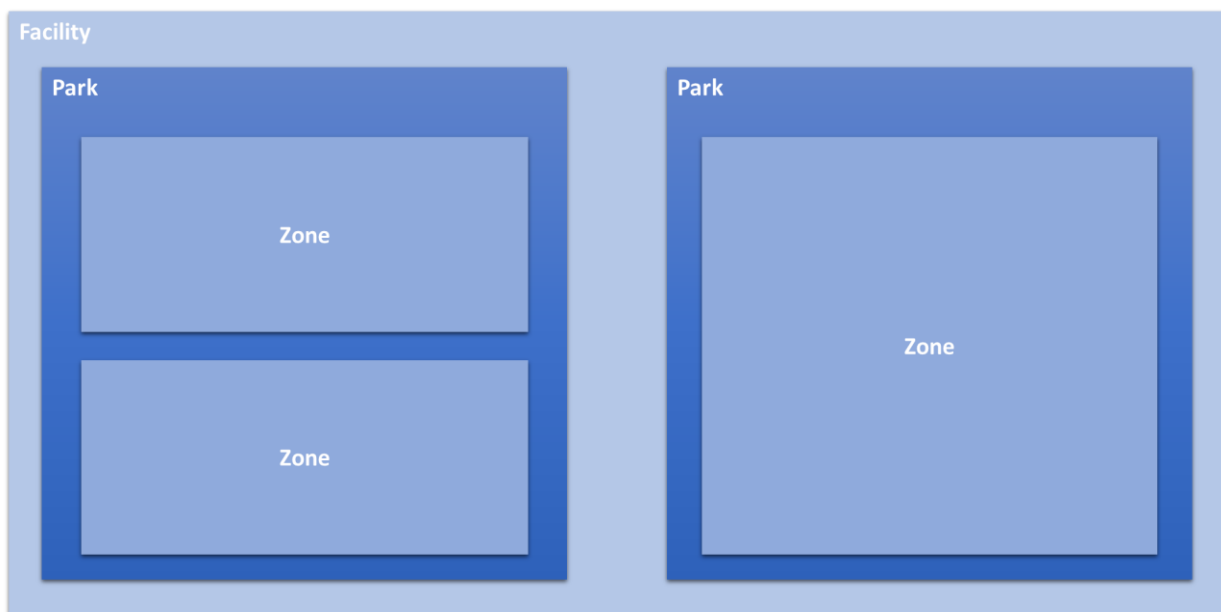
| Version | Amendment notes                                                     |
|---------|---------------------------------------------------------------------|
| 1.1     | Initial Documentation                                               |
| 2.0     | Added new API location attributes. New template.                    |
| 2.1     | Added new car park: Park&Ride – Emu Plains                          |
| 2.2     | Added new car park: Park&Ride – Riverwood                           |
| 2.3     | Added new car park: Park&Ride – North Rocks, Edmondson Park (north) |

# 1. Car Park API

The Car Park API provides real time and historical occupancy of selected car parks in NSW. This API provides the occupancy for Transport Park&Ride car parks. Transport Park&Ride is designed to free-up more spaces at commuter car parks for those who want to travel on public transport. These car parks are mapped to TfNSW Transit Stop Numbers (TSN) and Facility IDs. The data feed contains the parking occupancy information by type in real time.

An occupancy notification can be sent each time the car park facility logs a standard entry or exit transaction, or at a regular intervals. Refer to Section 4 Car Parks for update frequency.

The car parks can be configured differently. Below depicts an example of a car park facility configuration. Each facility can have one or more parks. Each park can have one or more zones.



## 1.1 Terms and Conditions

Transport for NSW provides access to the Car Park API with Car Park Occupancy on the following terms:

- The data is provided by outsource systems, which is outside the control of TfNSW.
- The occupancy data is an estimated count of vehicle in the car park as there are various reasons that may affect the count.
- TfNSW has provided some recommendations on representation of data to the customer to reduce confusion/interpretation by the customer.
- This API specification will be changed in the future to provision for other car park types and systems and aligned to TfNSW transmodel.
- Data may be reset at any time to handle errors caused by power outages, broken boom gates or any other issues that might cause operations issues.

## 2. Data Structure

### 2.1 High Level Data Explanation

#### 2.1.1 Facility information

A feed listing all the car park facilities will be returned if there is no facility ID provided in the request.

| Field Name    | Data Type | Field Description                                            |
|---------------|-----------|--------------------------------------------------------------|
| facility_id   | string    | The ID assigned to the facility.                             |
| facility_name | string    | The name given to the facility.<br>Prefixed with Park&Ride – |

#### 2.1.2 Car park occupancy

| Field Name | Data Type | Field Description                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| tsn        | string    | Transit Stop Number. A unique number identifying a stop or group of stops on the TfNSW network. It is made up of the post code of the suburb in which the transit stop is located and the next available sequential number. e.g. 2000001, 2000002. Note: transit stop numbers will not change – even if there is a change in the relevant postcode.                                                               |
| time       | string    | The time in the event. It is in seconds since year 2000.<br>Note: Please do not use this field. Instead, use MessageDate to retrieve the date and time of the feed.                                                                                                                                                                                                                                               |
| spots      | string    | Total number of parking spots in the car park.<br>Note: The number of spots does not include motorcycle and disabled bay.                                                                                                                                                                                                                                                                                         |
| zones      | array     | This contains the following: <ul style="list-style-type: none"><li>• ID for the zone</li><li>• Name for the zone</li><li>• Total number of parking spots in the zone</li><li>• Occupancy data for the zone</li><li>• ID for the parent zone</li></ul> Note: This information is not accurate for Ashfield, Kogarah, Seven Hills, Manly Vale and Brookvale car parks. However, it is accurate for other car parks. |
| parkID     | string    | This is a single digit static database identifier for the car park.                                                                                                                                                                                                                                                                                                                                               |
| location   | object    | Location broken down by type.<br>Refer to location object table below.                                                                                                                                                                                                                                                                                                                                            |

| Field Name         | Data Type | Field Description                                                                                                                                                           |
|--------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| occupancy          | object    | Occupancy broken down by type.<br>Refer to occupancy object table below.                                                                                                    |
| MessageDate        | string    | Date and time of the event.                                                                                                                                                 |
| facility_id        | string    | The ID assigned to the car park facility.                                                                                                                                   |
| facility_name      | string    | The name given to the car park facility.                                                                                                                                    |
| tfnswh_facility_id | string    | This ID is used by TfNSW to identify the car park facility. The format is <TSN><Carpark Type><id>.<br>Carpark Type:<br>TPR - Transport Park&Ride<br>CCP - Commuter Car Park |

### 2.1.3 Location

| Field Name | Data Type | Field Description                            |
|------------|-----------|----------------------------------------------|
| suburb     | String    | The suburb where the car park is located     |
| address    | string    | The street address of the car park           |
| latitude   | string    | The latitude co-ordinate for car park entry  |
| longitude  | string    | The longitude co-ordinate for car park entry |

### 2.1.4 Occupancy

| Field Name | Data Type | Field Description                                                                                                                                                     |
|------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| loop       | string    | This is the counter for the loops being armed or disarmed.                                                                                                            |
| total      | string    | This is the counter information for the number of vehicles in the car park.<br>Note: Use this to get an accurate estimates of the number of vehicles in the car park. |
| monthlies  | string    | These are the number of vehicles in the car park using monthlies or monthly parking scheme. This is not used in this implementation.                                  |
| open_gate  | string    | This is information when the car park operator manually opens the gate. This is not used in this implementation.                                                      |
| transients | string    | This is the counter for the number of vehicles with active transactions in the system.<br>Note: Manual opening of gantries does not decrease this count.              |

## 2.1.5 Sample feed

```
{
  "tsn": "2155384",
  "time": "785808515",
  "spots": "123",
  "zones": [
    {
      "spots": "123",
      "zone_id": "1",
      "occupancy": {
        "loop": "4786",
        "total": "123",
        "monthlies": null,
        "open_gate": null,
        "transients": null
      },
      "zone_name": "SYD396A Tallawong P1 Car Park",
      "parent_zone_id": "0"
    }
  ],
  "ParkID": "1",
  "location": {
    "suburb": "Tallawong",
    "address": "Conferta Avenue",
    "latitude": "-33.69304704",
    "longitude": "150.9052577"
  },
  "occupancy": {
    "loop": "4786",
    "total": "123",
    "monthlies": null,
    "open_gate": null,
    "transients": null
  },
  "MessageDate": "2024-11-25T11:08:35",
  "facility_id": "26",
  "facility_name": "Park&Ride - Tallawong P1",
  "tfnsf_facility_id": "2155384TPR001"
}
```

### 3. Recommendation

Transport for NSW recommends the following implementation:

1. To calculate the number of available parking lots, it is recommended to use the following formula.  
$$\text{Availability} = \text{spots} - \text{total}$$
2. Displays the number of parking lots left in the car park facility. Use the “total” in facility occupancy object for an accurate estimates of occupancy information of the car park.
3. Calculate and display “Full” when the available parking lots drop below 1.
4. Display “Almost Full” when the available parking lots drop to less than 10%. This figure should be configurable and adjusted according to customers’ feedback.
5. To manage customers’ expectations, indicate that the car park data may differ from actual parking. It is recommended to put that the approximate sign (~) to inform the customers that this is an estimated car park availability information.

### 4. Car Parks

| ID | Name                                                       | TSN     | TSN Coordinates          | Number of spots | Update Frequency                  |
|----|------------------------------------------------------------|---------|--------------------------|-----------------|-----------------------------------|
| 1  | Tallawong Station Car Park <b>(Historical Only)</b>        | 2155384 | -33.69163<br>150.906022  | 1004            | 60s                               |
| 2  | Kellyville Station Car Park <b>(Historical Only)</b>       | 2155382 | -33.713514<br>150.935304 | 1374            | 60s                               |
| 3  | Bella Vista Station Car Park <b>(Historical Only)</b>      | 2153478 | -33.730592<br>150.944024 | 800             | 60s                               |
| 4  | Hills Showground Station Car Park <b>(Historical Only)</b> | 2154392 | -33.72782<br>150.987345  | 600             | 60s                               |
| 5  | Cherrybrook Station Car Park <b>(Historical Only)</b>      | 2126158 | -33.736703<br>151.031977 | 400             | 60s                               |
| 6  | Park&Ride - Gordon Henry St (north)                        | 207210  | -33.756009<br>151.154528 | 213             | 15s                               |
| 7  | Park&Ride - Kiama                                          | 253330  | -34.672518<br>150.854695 | 42              | 15s                               |
| 8  | Park&Ride - Gosford                                        | 225040  | -33.423883<br>151.341711 | 1057            | On counter change                 |
| 9  | Park&Ride - Revesby                                        | 221210  | -33.95246<br>151.014838  | 934             | On counter change or every 10mins |
| 10 | Park&Ride - Warriewood                                     | 2101131 | 33.697777<br>151.300667  | 244             | 15s                               |
| 11 | Park&Ride - Narrabeen                                      | 2101130 | -33.713514<br>151.297315 | 46              | On counter change or every 10mins |
| 12 | Park&Ride - Mona Vale                                      | 2103108 | -33.677276<br>151.305146 | 68              | 15s                               |



| ID | Name                                       | TSN     | TSN Coordinates          | Number of spots | Update Frequency                  |
|----|--------------------------------------------|---------|--------------------------|-----------------|-----------------------------------|
| 13 | Park&Ride - Dee Why                        | 2099207 | -33.752797<br>151.286485 | 117             | On counter change or every 10mins |
| 14 | Park&Ride - West Ryde                      | 211420  | -33.807172<br>151.090229 | 151             | On counter change or every 10mins |
| 15 | Park&Ride - Sutherland East Parade         | 223210  | -34.031787<br>151.05719  | 373             | On counter change                 |
| 16 | Park&Ride - Leppington                     | 217933  | -33.9544<br>150.8081     | 1884            | On counter change or every 10mins |
| 17 | Park&Ride - Edmondson Park (south)         | 217426  | -33.9693<br>150.8587     | 1431            | On counter change or every 10mins |
| 18 | Park&Ride - St Marys                       | 276010  | -33.762256<br>150.776029 | 682             | On counter change                 |
| 19 | Park&Ride - Campbelltown Farrow Rd (north) | 256020  | -34.063835<br>150.813929 | 68              | On counter change or every 10mins |
| 20 | Park&Ride - Campbelltown Hurley St         | 256020  | -34.063835<br>150.813929 | 118             | On counter change or every 10mins |
| 21 | Park&Ride - Penrith (at-grade)             | 275010  | -33.750055<br>150.696135 | 230             | On counter change or every 10mins |
| 22 | Park&Ride - Penrith (multi-level)          | 275010  | -33.750055<br>150.696135 | 1144            | On counter change or every 10mins |
| 23 | Park&Ride - Warwick Farm                   | 217010  | -33.91345<br>150.935036  | 910             | On counter change                 |
| 24 | Park&Ride - Schofields                     | 276220  | -33.704477<br>150.873817 | 700             | On counter change or every 10mins |
| 25 | Park&Ride - Hornsby                        | 207720  | -33.702801<br>151.098494 | 145             | On counter change or every 10mins |
| 26 | Park&Ride - Tallawong P1                   | 2155384 | -33.69163<br>150.906022  | 121             | On counter change or every 10mins |
| 27 | Park&Ride - Tallawong P2                   | 2155384 | -33.69163<br>150.906022  | 455             | On counter change or every 10mins |
| 28 | Park&Ride - Tallawong P3                   | 2155384 | -33.69163<br>150.906022  | 397             | On counter change or every 10mins |
| 29 | Park&Ride - Kellyville (north)             | 2155382 | -33.713514<br>150.935304 | 351             | On counter change or every 10mins |
| 30 | Park&Ride - Kellyville (south)             | 2155382 | -33.713514<br>150.935304 | 964             | On counter change or every 10mins |
| 31 | Park&Ride - Bella Vista                    | 2153478 | -33.730592<br>150.944024 | 777             | On counter change or every 10mins |
| 32 | Park&Ride - Hills Showground               | 2154392 | -33.72782<br>150.987345  | 584             | On counter change or every 10mins |
| 33 | Park&Ride - Cherrybrook                    | 2126158 | -33.736703<br>151.031977 | 384             | On counter change or every 10mins |

| ID  | Name                                | TSN                          | TSN Coordinates                 | Number of spots | Update Frequency                  |
|-----|-------------------------------------|------------------------------|---------------------------------|-----------------|-----------------------------------|
| 34  | Park&Ride – Lindfield Village Green | 207010                       | -33.775185<br>151.169111        | 94              | On counter change or every 10mins |
| 35  | Park&Ride – Beverly Hills           | 220910                       | -33.948849<br>151.081692        | 200             | On counter change or every 10mins |
| 36  | Park&Ride – Emu Plains              | 275020                       | -33.745527<br>150.66987         | 751             | On counter change or every 10mins |
| 37  | Park&Ride – Riverwood               | 221010                       | -33.952727<br>151.050035        | 142             | On counter change or every 10mins |
| 38  | Park&Ride – North Rocks             | 2151161                      | -33.765539<br>151.014131        | 139             | On counter change or every 10mins |
| 39  | Park&Ride – Edmondson Park (north)  | 217426                       | -33.969123<br>150.861594        | 917             | On counter change or every 10mins |
| 486 | Park&Ride – Ashfield                | 213110                       | -33.8875506079<br>151.125504163 | 225             | On counter change or every 10mins |
| 487 | Park&Ride – Kogarah                 | 221710                       | -33.9621493059<br>151.132641462 | 259             | On counter change or every 10mins |
| 488 | Park&Ride – Seven Hills             | 214710                       | -33.774430434<br>150.936513359  | 1613            | On counter change or every 10mins |
| 489 | Park&Ride – Manly Vale              | (Parent Stop TSN)<br>2093117 | -33.786247<br>151.26671         | 142             | On counter change or every 10mins |
| 490 | Park&Ride – Brookvale               | 210020                       | -33.767508<br>151.268541        | 246             | On counter change or every 10mins |

Notes:

- A parent stop number 2093117 was created for Manly Vale stops 209325 and 209326. This was created as it is not associated with a station, ferry wharf or light rail stop. Please add this parent stop manually where required as it is not in the GTFS feed.
- The number of spots provided in the table are for quick reference. Do not hard code these figures as they may change from time to time. You can get the number of spots for each car park in the feed.
- There are works expected at Kiama which may intermittently impact the real time parking availability.
- Location information was added to the car park API in July 2024. Latitude and Longitude co-ordinates in the API indicate the location of car park entry.